

Problem 1

1. Numerical Integration: Assume a C^1 Euler-Bernoulli Beam element (with Hermitian shape functions) and the shown nodal displacements. Assume a nonlinear moment-curvature constitutive relation for the beam as shown. Use numerical integration rules below and

Step 1: Find the element nodal force vector using $\mathbf{P} = \int_{-1}^1 \mathbf{B}^T M(\kappa) d\xi$. Draw the BMD.

Step 2: Find the 4x4 element tangent stiffness matrix using $\mathbf{K}_T = \int_{-1}^1 \mathbf{B}^T \frac{dM}{d\kappa} \mathbf{B} d\xi$.

Compare the results between the different cases and state your observations with reason.

- a. Gaussian quadrature with 3 and 5 integration points.
- b. Gauss-Lobatto with 3, 5, and 7 integration points.

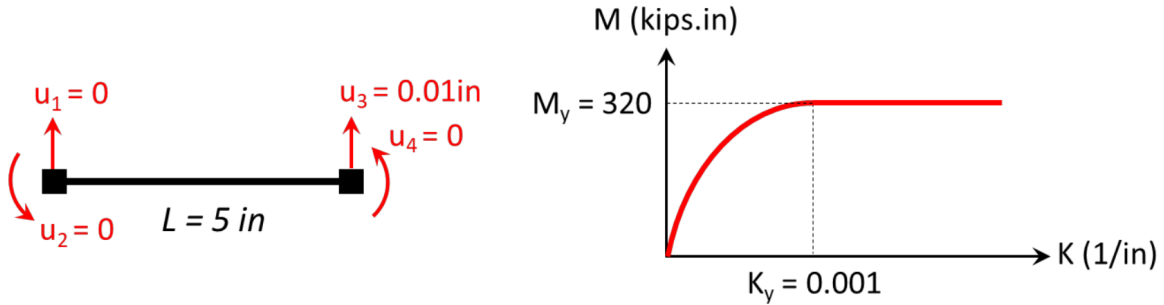


Figure 1: Numerical stiffness matrix output.

$$M(\xi) = \begin{cases} 320, & \xi \in [-1, -5/12] \\ -768\xi, & \xi \in [-5/12, 5/12] \\ -320, & \xi \in [5/12, 1] \end{cases}$$

$$\frac{dM}{d\kappa} = \begin{cases} 0, & \kappa \in [-1, -5/12] \\ EI = 320,000, & \kappa \in [-5/12, 5/12] \\ 0, & \kappa \in [5/12, 1] \end{cases}$$

$$\mathbf{P} = 2.5 \int_{-1}^1 \mathbf{B}^T(\xi) M(\xi) d\xi$$

$$P = 2.5 \left[\int_{-1}^{-5/12} B^T(\xi) M(\xi) d\xi + \int_{-5/12}^{5/12} B^T(\xi) M(\xi) d\xi + \int_{5/12}^1 B^T(\xi) M(\xi) d\xi \right]$$

$$\mathbf{P}_{\text{analytical}} = \begin{bmatrix} -181.44 \\ -451.68 \\ 181.44 \\ -481.68 \end{bmatrix}$$

$$K_T = 2.5 \int_{-1}^1 \mathbf{B}^T \frac{dM}{d\kappa} \mathbf{B} d\xi$$

$$\mathbf{K}_T = 800,000 \int_{-5/12}^{5/12} \begin{bmatrix} 0.24\xi \\ 0.6\xi - 0.2 \\ -0.24\xi \\ 0.6\xi + 0.2 \end{bmatrix} [0.24\xi \quad 0.6\xi - 0.2 \quad -0.24\xi \quad 0.6\xi + 0.2] d\xi$$

$$\mathbf{K}_{\text{analytical}} = \begin{bmatrix} 2222.22 & 5555.56 & -2222.22 & 5555.56 \\ 5555.56 & 40555.56 & -5555.56 & -12777.78 \\ -2222.22 & -5555.56 & 2222.22 & -5555.56 \\ 5555.56 & -12777.78 & -5555.56 & 40555.56 \end{bmatrix}$$

Gauss-Quadrature Results

Python Code: https://github.com/nirajkyadav/Homeworks/blob/main/HW4/Q1_gauss_legendre_stiffness.py

3 Points

$$\mathbf{K}_{\text{GQ},3} = \begin{bmatrix} 2222.22 & 5555.56 & -2222.22 & 5555.56 \\ 5555.56 & 40555.56 & -5555.56 & -12777.78 \\ -2222.22 & -5555.56 & 2222.22 & -5555.56 \\ 5555.56 & -12777.78 & -5555.56 & 40555.56 \end{bmatrix}$$

$$\max |\mathbf{K}_{3\text{pt}} - \mathbf{K}_{\text{analytical}}| = 0$$

5 Points

$$\mathbf{K}_{\text{GQ},5} = \begin{bmatrix} 2222.22 & 5555.56 & -2222.22 & 5555.56 \\ 5555.56 & 40555.56 & -5555.56 & -12777.78 \\ -2222.22 & -5555.56 & 2222.22 & -5555.56 \\ 5555.56 & -12777.78 & -5555.56 & 40555.56 \end{bmatrix}$$

$$\max |\mathbf{K}_{5\text{pt}} - \mathbf{K}_{\text{analytical}}| = 0$$

Observation

Both the 3-point and 5-point Gauss-Quadrature rules yield the same stiffness matrix, matching the analytical solution. This occurs because the strain-displacement matrix $\mathbf{B}(\xi)$ has linear dependence on ξ , which makes the product $\mathbf{B}^T(\xi)\mathbf{B}(\xi)$ at most a quadratic polynomial in ξ .

An n -point Gauss-Quadrature rule integrates exactly all polynomials up to degree $2n - 1$. Therefore, the 3-point rule is exact for polynomials up to degree 5, and the 5-point rule up to degree 9.

Since the integrand in this case is only quadratic (degree 2), even the 3-point rule already provides exact integration. Increasing the number of integration points to 5 does not improve accuracy, as the integral is already evaluated exactly.

Similarly, adopting a segmented (continuous in piecewise) integration approach for the internal nodal forces by evaluating each zone individually would also yield mathematically exact results.

Gauss–Lobatto Results

Python Code: https://github.com/nirajkyadav/Homeworks/blob/main/HW4/Q1_gauss_lobatto_stiffness.py

3 Points

$$\mathbf{K}_{\text{GL},3} = \begin{bmatrix} 2222.22 & 5555.56 & -2222.22 & 5555.56 \\ 5555.56 & 40555.56 & -5555.56 & -12777.78 \\ -2222.22 & -5555.56 & 2222.22 & -5555.56 \\ 5555.56 & -12777.78 & -5555.56 & 40555.56 \end{bmatrix}$$
$$\max |\mathbf{K}_{\text{GL},3} - \mathbf{K}_{\text{analytical}}| = 0$$

5 Points

$$\mathbf{K}_{\text{GL},5} = \begin{bmatrix} 2222.22 & 5555.56 & -2222.22 & 5555.56 \\ 5555.56 & 40555.56 & -5555.56 & -12777.78 \\ -2222.22 & -5555.56 & 2222.22 & -5555.56 \\ 5555.56 & -12777.78 & -5555.56 & 40555.56 \end{bmatrix}$$
$$\max |\mathbf{K}_{\text{GL},5} - \mathbf{K}_{\text{analytical}}| = 0$$

7 Points

$$\mathbf{K}_{\text{GL},7} = \begin{bmatrix} 2222.22 & 5555.56 & -2222.22 & 5555.56 \\ 5555.56 & 40555.56 & -5555.56 & -12777.78 \\ -2222.22 & -5555.56 & 2222.22 & -5555.56 \\ 5555.56 & -12777.78 & -5555.56 & 40555.56 \end{bmatrix}$$
$$\max |\mathbf{K}_{\text{GL},7} - \mathbf{K}_{\text{analytical}}| = 0$$

Observation

All 3-point, 5-point, and 7-point Gauss–Lobatto rules yield the same stiffness matrix, matching the analytical solution.

An n -point Gauss–Lobatto rule integrates exactly all polynomials up to degree $2n - 3$. Therefore, the 3-point rule is exact for polynomials up to degree 3, and the 5-point rule up to degree 7.

Since the integrand in this case is only quadratic (degree 2), even the 3-point rule already provides exact integration. Increasing the number of integration points to 5, and 7 does not improve accuracy, as the integral is already evaluated exactly.

Similarly, adopting a segmented (continuous in piecewise) integration approach for the internal nodal forces by evaluating each zone individually would also yield mathematically exact results.